KCE PATHWAYS · GRADE 3 · UNIT 3: MARKET DAYS · WEEK 12 · MATH

Sharing at the Chicago Bakery — Teacher Lesson Plan & Answer Key

Learning objectives: Students understand division as sharing into equal groups or making equal groups, fluently divide by 2 through 9 by thinking of the related multiplication fact, and write fact families that connect multiplication and division. They find the unknown number in a multiplication or division fact (for example 6 × ? = 42 or 48 ÷ ? = 8) and solve one- and two-step word problems.

Duration: 35–45 minutes
Materials: Computer or tablet with the lesson open · pencil · printed Pre & Post sheet

Lesson flow (as it appears on screen)

1. Title and name: student types first and last name so results go to KingCretot Experience.
2. Warm-Up (Week 11 review, 15 pts): 6s–9s multiplication facts, turn-around, break-apart, 9s pattern, doubling, and the identity and zero properties. These facts are the key to today's division, so if a student misses two of three, quickly review the 6s–9s facts first.
3. Picture It: A Morning at a Chicago Bakery. Meets Lena, Kofi, Babcia, Mr. Brooks, and Diego at the Nowak Family Bakery, plus Chicago facts (Lake Michigan, the L and the Loop, and paczki). Stress the big idea: sharing into equal groups is division.
4. Word Wrangler: 6 vocab cards (divide, dividend, divisor, quotient, fact family, unknown). Mention that multiplication and division are inverse operations.
5. Pencil Break (Pre): student completes the Pre side of the printed sheet before any teaching. Do not help yet; it shows what they already know.
6. Sharing Into Equal Boxes (10 pts). Tip: act it out with 12–24 small objects and cups. Show both kinds of division: how many in each group, and how many groups.
7. Fact Families: One Family, Four Facts (10 pts). Tip: draw a fact family house with the three numbers in the roof and the four facts inside.
8. Think Multiplication: Dividing by 6, 7, 8, and 9 (10 pts). Tip: have the student say the question aloud, for example "6 times what number equals 48?" Use a multiplication chart if needed.
9. Mystery Numbers: Finding the Unknown (15 pts). Tip: if the unknown is a factor, divide; if the unknown is the starting number in a division fact, multiply. Always check by putting the number back into the fact.
10. Bonus Practice: Loading the Delivery Bike (XP only).
11. Match-Up Round (10 pts, 2 per first-try match).
12. Fix the Sentence (10 pts): typed answers are checked exactly (capital letter, end mark, list commas). Both serial-comma and no-serial-comma list forms are accepted.
13. Speed Arena and Keyboard Corral (XP only).
14. Boss Battle: The Big Bakery Order (20 pts): two-step problem, which-explanation-is-correct, unknown or word problem, find-the-mistake.
15. Pencil Break (Post): student completes the Post side of the sheet and the star challenge if time allows.

16. Results: score of 85 or higher shows a custom Bakery Hero congratulations with confetti; below 85 shows encouragement and a fresh question set on the next try (sets rotate 1, 2, 3). Both remind the student to finish the Post-Lesson Challenge sheet.

Lesson answer key with reasoning

Every scored item the student sees, in order, across all three rotating question sets. The "why" is the same reasoning shown on screen after a second missed try.

WARM-UP: WEEK 11 REVIEW

ID	Question	Answer	Why (explain to student)
w1_s0	Set 1: Remember Rosa's orange grove? Rosa packed 8 crates with 9 oranges in each crate. How many oranges?	**72**	8 × 9 = 72 oranges. Check with the 9s pattern: 7 + 2 = 9.
w2_s0	Set 1: Rosa knows 6 × 8 = 48. Which fact must also equal 48?	**8 × 6**	The turn-around property says switching the factors keeps the same product, so 8 × 6 = 48.
w3_s0	Set 1: Use the break-apart trick. 7 × 9 = 7 × 5 + 7 × 4 = 35 + 28 = ___	**63**	35 + 28 = 63, so 7 × 9 = 63.
w1_s1	Set 2: A block of the grove has 6 rows with 7 trees in each row. How many trees?	**42**	6 × 7 = 42 trees.
w2_s1	Set 2: Which fact is true?	**8 × 0 = 0**	Any number times 0 is 0, so 8 × 0 = 0. (And 8 × 1 = 8.)
w3_s1	Set 2: Use the 9s pattern. 9 × 6 = ___	**54**	The tens digit is one less than 6, which is 5, and 5 + 4 = 9. So 9 × 6 = 54.
w1_s2	Set 3: Each crate holds 8 oranges. How many oranges are in 8 crates?	**64**	8 × 8 = 64 oranges.
w2_s2	Set 3: Which property says 9 × 1 = 9?	**identity property**	The identity property says any number times 1 stays the same.
w3_s2	Set 3: Double a 4s fact to find an 8s fact. 4 × 7 = 28, so 8 × 7 = ___	**56**	Double 28 is 56, so 8 × 7 = 56.

SHARING INTO EQUAL BOXES

ID	Question	Answer	Why (explain to student)
d1a_s0	Set 1: Lena packs 12 bagels into 2 bags with the same number in each bag. How many bagels go in each bag?	**6**	12 ÷ 2 = 6 bagels in each bag, because 2 × 6 = 12.
d1b_s0	Set 1: Which explanation is correct? Kofi has 20 cookies and puts 5 cookies in each bag. Kofi says he can fill 4 bags. Lena says he can fill 15 bags because 20 − 5 = 15.	**Kofi is correct.**	20 ÷ 5 = 4, because 4 groups of 5 make 20. Lena only took away one group of 5.
d1a_s1	Set 2: Mr. Brooks puts 18 rolls on 3 trays with the same number on each tray. How many rolls are on each tray?	**6**	18 ÷ 3 = 6 rolls on each tray, because 3 × 6 = 18.
d1b_s1	Set 2: Find the mistake. Lena wrote 16 ÷ 4 = 12. What is wrong?	**She subtracted. 16 ÷ 4 = 4, because 4 × 4 = 16.**	Lena found 16 − 4 = 12. To divide, find how many 4s make 16: 4 × 4 = 16, so 16 ÷ 4 = 4.

ID	Question	Answer	Why (explain to student)
d1a_s2	Set 3: Kofi has 20 muffins. He puts 4 muffins in each box. How many boxes does he fill?	**5**	20 ÷ 4 = 5 boxes, because 5 × 4 = 20.
d1b_s2	Set 3: 30 pierogi are shared equally on 5 plates. Which fact shows how many pierogi go on each plate?	**30 ÷ 5 = 6**	Sharing 30 into 5 equal groups is division: 30 ÷ 5 = 6 pierogi on each plate.

FACT FAMILIES: ONE FAMILY, FOUR FACTS

ID	Question	Answer	Why (explain to student)
f1a_s0	Set 1: Which fact is in the same fact family as 7 × 8 = 56?	**56 ÷ 7 = 8**	56 ÷ 7 = 8 uses the same three numbers: 7, 8, and 56.
f1b_s0	Set 1: Find the mistake. Kofi wrote this fact family for 3, 9, and 27: 3 × 9 = 27, 9 × 3 = 27, 27 ÷ 3 = 9, 27 ÷ 9 = 4. Which fact is wrong?	**27 ÷ 9 = 4 is wrong. It should be 27 ÷ 9 = 3.**	Every fact in the family uses 3, 9, and 27. Since 9 × 3 = 27, 27 ÷ 9 = 3.
f1a_s1	Set 2: Use the fact family. 8 × 6 = 48, so 48 ÷ 6 = ___	**8**	6, 8, and 48 are one fact family. 6 × 8 = 48, so 48 ÷ 6 = 8.
f1b_s1	Set 2: Which explanation is correct? Babcia asks: why is 36 ÷ 4 equal to 9?	**Because 4 × 9 = 36, and division undoes multiplication.**	4, 9, and 36 are a fact family. 4 × 9 = 36, so 36 ÷ 4 = 9.
f1a_s2	Set 3: Which three numbers make a multiplication and division fact family?	**2, 9, 18**	2 × 9 = 18, so 2, 9, and 18 make the facts 2 × 9, 9 × 2, 18 ÷ 2, and 18 ÷ 9.
f1b_s2	Set 3: Find the mistake. Lena wrote: 8 × 9 = 72, so 72 ÷ 8 = 8. What is wrong?	**72 ÷ 8 should be 9, because 8 × 9 = 72.**	The fact family is 8, 9, and 72. If 8 × 9 = 72, then 72 ÷ 8 = 9.

THINK MULTIPLICATION: DIVIDING BY 6, 7, 8, AND 9

ID	Question	Answer	Why (explain to student)
f2a_s0	Set 1: Diego has 35 bagels to deliver in bags of 7. How many bags does he need?	**5**	Think 7 × ? = 35. Since 7 × 5 = 35, 35 ÷ 7 = 5 bags.
f2b_s0	Set 1: Which explanation is correct? How can you find 72 ÷ 8?	**Think 8 × ? = 72. Since 8 × 9 = 72, the answer is 9.**	Division undoes multiplication. 8 × 9 = 72, so 72 ÷ 8 = 9.
f2a_s1	Set 2: Babcia makes 56 pierogi and puts 8 pierogi on each plate. How many plates does she fill?	**7**	Think 8 × ? = 56. Since 8 × 7 = 56, 56 ÷ 8 = 7 plates.
f2b_s1	Set 2: Find the mistake. Kofi says 36 ÷ 6 = 7 because 6 × 7 = 36. What is wrong?	**6 × 7 is 42, not 36. Since 6 × 6 = 36, 36 ÷ 6 = 6.**	Kofi used a wrong multiplication fact. 6 × 6 = 36, so 36 ÷ 6 = 6.
f2a_s2	Set 3: Mr. Brooks bakes 81 rolls and puts 9 rolls in each bag. How many bags does he fill?	**9**	Think 9 × ? = 81. Since 9 × 9 = 81, 81 ÷ 9 = 9 bags.
f2b_s2	Set 3: Which multiplication fact helps you find 49 ÷ 7?	**7 × 7 = 49**	7 × 7 = 49, so 49 ÷ 7 = 7.

MYSTERY NUMBERS: FINDING THE UNKNOWN

ID	Question	Answer	Why (explain to student)
u1a_s0	Set 1: Find the unknown number. 6 × ? = 42	**7**	6 × 7 = 42, so the unknown is 7. You can also divide: 42 ÷ 6 = 7.

ID	Question	Answer	Why (explain to student)
u1b_s0	Set 1: Find the unknown number. 40 ÷ ? = 8	**5**	40 divided by what number is 8? Since 8 × 5 = 40, the unknown is 5.
u1c_s0	Set 1: Find the mistake. Kofi solved ? ÷ 4 = 7 and wrote 3. What went wrong?	**He subtracted. The unknown is 28, because 4 × 7 = 28.**	When the starting number is unknown, multiply: 4 × 7 = 28. Check: 28 ÷ 4 = 7.
u1a_s1	Set 2: Find the unknown number. 6 × ? = 30	**5**	6 × 5 = 30, so the unknown is 5. You can also divide: 30 ÷ 6 = 5.
u1b_s1	Set 2: Find the unknown number. ? ÷ 3 = 9	**27**	The starting number is unknown, so multiply: 3 × 9 = 27. Check: 27 ÷ 3 = 9.
u1c_s1	Set 2: Which explanation is correct? To solve 9 × ? = 81, Lena says to divide: 81 ÷ 9 = 9. Kofi says to subtract: 81 − 9 = 72.	**Lena is correct.**	A missing factor is found by dividing. 81 ÷ 9 = 9, and 9 × 9 = 81 checks it.
u1a_s2	Set 3: Find the unknown number. ? × 6 = 36	**6**	6 × 6 = 36, so the unknown is 6. You can also divide: 36 ÷ 6 = 6.
u1b_s2	Set 3: Find the unknown number. 35 ÷ ? = 5	**7**	35 divided by what number is 5? Since 5 × 7 = 35, the unknown is 7.
u1c_s2	Set 3: Diego puts 6 paczki in each box and fills 7 boxes. Which fact matches the story, if the unknown is how many paczki he packed in all?	**? ÷ 6 = 7**	The total is split into boxes of 6 and makes 7 boxes, so the total divided by 6 is 7. The unknown is 42, because 6 × 7 = 42.

MATCH-UP ROUND (2 POINTS PER FIRST-TRY MATCH)

ID	Question	Answer	Why (explain to student)
match_s0_1	Set 1: Match 30 ÷ 5	**6**	2 points if matched on the first try.
match_s0_2	Set 1: Match 16 ÷ 2	**8**	2 points if matched on the first try.
match_s0_3	Set 1: Match 81 ÷ 9	**9**	2 points if matched on the first try.
match_s0_4	Set 1: Match 25 ÷ 5	**5**	2 points if matched on the first try.
match_s0_5	Set 1: Match 49 ÷ 7	**7**	2 points if matched on the first try.
match_s1_1	Set 2: Match quotient	**the answer to a division problem**	2 points if matched on the first try.
match_s1_2	Set 2: Match divisor	**the number you divide by**	2 points if matched on the first try.
match_s1_3	Set 2: Match dividend	**the total you start with and split up**	2 points if matched on the first try.
match_s1_4	Set 2: Match fact family	**facts that use the same three numbers**	2 points if matched on the first try.
match_s1_5	Set 2: Match 5 × ? = 35	**7**	2 points if matched on the first try.
match_s2_1	Set 3: Match 2 × ? = 14	**7**	2 points if matched on the first try.
match_s2_2	Set 3: Match ? ÷ 3 = 6	**18**	2 points if matched on the first try.
match_s2_3	Set 3: Match 36 ÷ ? = 9	**4**	2 points if matched on the first try.
match_s2_4	Set 3: Match 3 × ? = 27	**9**	2 points if matched on the first try.
match_s2_5	Set 3: Match 16 ÷ 2	**8**	2 points if matched on the first try.

FIX THE SENTENCE

ID	Question	Answer	Why (explain to student)
fs1_s0	Set 1: Fix the sentence: lena packed 6 boxes of rolls	**Lena packed 6 boxes of rolls.**	Lena is a name and starts the sentence, so it gets a capital L. A telling sentence ends with a period.
fs2_s0	Set 1: Fix the sentence: we sold bagels muffins and rolls	**We sold bagels, muffins, and rolls. (also accepted: We sold bagels, muffins and rolls.)**	Start with a capital W, put commas between the things in the list, and end with a period.
fs1_s1	Set 2: Fix the sentence: the train goes over the street	**The train goes over the street.**	Start the sentence with a capital T, and end the telling sentence with a period.
fs2_s1	Set 2: Fix the sentence: kofi lena and diego shared the paczki	**Kofi, Lena, and Diego shared the paczki. (also accepted: Kofi, Lena and Diego shared the paczki.)**	Each name gets a capital letter, commas go between the names in the list, and the sentence ends with a period.
fs1_s2	Set 3: Fix the sentence: how many muffins are in each box	**How many muffins are in each box?**	Start with a capital H. This sentence asks something, so it ends with a question mark.
fs2_s2	Set 3: Fix the sentence: babcia made rolls cookies and pierogi	**Babcia made rolls, cookies, and pierogi. (also accepted: Babcia made rolls, cookies and pierogi.)**	Babcia starts the sentence and is what Lena calls her grandma, so it gets a capital B. Put commas between the things in the list and end with a period.

BOSS BATTLE: THE BIG BAKERY ORDER

ID	Question	Answer	Why (explain to student)
bb1_s0	Set 1: Two steps! Babcia bakes 4 trays with 9 paczki on each tray. She packs all of them into boxes of 6. How many boxes does she fill?	**6**	Step 1: 4 × 9 = 36 paczki. Step 2: 36 ÷ 6 = 6 boxes.
bb2_s0	Set 1: Which explanation is correct? Lena found 64 ÷ 8 by thinking 8 × 8 = 64. Kofi found it by thinking 64 − 8 = 56.	**Lena is correct.**	Lena used a fact family: 8 × 8 = 64, so 64 ÷ 8 = 8. Kofi subtracted one group of 8 instead of dividing.
bb3_s0	Set 1: Find the unknown number. 6 × ? = 54	**9**	6 × 9 = 54, so the unknown is 9. Check: 54 ÷ 6 = 9.
bb4_s0	Set 1: Find the mistake. Diego wrote this fact family: 3 × 8 = 24, 8 × 3 = 24, 24 ÷ 3 = 8, 24 ÷ 8 = 8. Which fact is wrong?	**24 ÷ 8 = 8 is wrong. It should be 24 ÷ 8 = 3.**	The family uses 3, 8, and 24. Since 8 × 3 = 24, 24 ÷ 8 = 3.
bb1_s1	Set 2: Two steps! Mr. Brooks baked 50 rolls. Five rolls were too small to sell. He packs the rest into bags of 9. How many bags does he fill?	**5**	Step 1: 50 − 5 = 45 rolls. Step 2: 45 ÷ 9 = 5 bags.
bb2_s1	Set 2: Find the mistake. Lena says 32 ÷ 4 = 9 because 4 × 9 = 32. What is wrong?	**4 × 9 is 36. Since 4 × 8 = 32, 32 ÷ 4 = 8.**	Lena used a wrong fact. 4 × 8 = 32, so 32 ÷ 4 = 8.
bb3_s1	Set 2: Diego rides past the L station with 72 muffins packed in boxes of 9. How many boxes does he have?	**8**	72 ÷ 9 = 8 boxes, because 9 × 8 = 72.

ID	Question	Answer	Why (explain to student)
bb4_s1	Set 2: Which explanation is correct? Why are 8 × 3 = 24 and 24 ÷ 3 = 8 in the same fact family?	**They use the same three numbers, 3, 8, and 24, and division undoes multiplication.**	A fact family uses the same three numbers. 8 × 3 = 24, and dividing 24 by 3 brings you back to 8.
bb1_s2	Set 3: Two steps! Kofi has 3 bags with 8 cookies in each bag. He shares all the cookies equally among 6 friends. How many cookies does each friend get?	**4**	Step 1: 3 × 8 = 24 cookies. Step 2: 24 ÷ 6 = 4 cookies for each friend.
bb2_s2	Set 3: Which explanation is correct? To find the unknown in ? ÷ 8 = 5, Lena multiplies 8 × 5 = 40. Kofi says the unknown is 3, because 8 − 5 = 3.	**Lena is correct.**	The starting number is unknown, so multiply: 8 × 5 = 40. Check: 40 ÷ 8 = 5.
bb3_s2	Set 3: Mr. Brooks puts 45 rolls on a tray in 5 equal rows. How many rolls are in each row?	**9**	45 ÷ 5 = 9 rolls in each row, because 5 × 9 = 45.
bb4_s2	Set 3: Find the mistake. Diego solved 7 × ? = 21 and said the unknown is 14. What is wrong?	**He subtracted 7 from 21. The unknown is 3, because 7 × 3 = 21.**	A missing factor is found by dividing: 21 ÷ 7 = 3. Check: 7 × 3 = 21.

Pre & Post sheet answer guide

Many items are open-ended. Grade for genuine understanding and complete sentences, not exact wording.

PRE-LESSON WARM-UP

#	Prompt	Model answer / grading note
1	7 crates of 8 oranges	7 × 8 = 56 oranges (8 × 7 = 56 also accepted).
2	12 rolls shared into 3 boxes	Drawing shows 3 boxes with 4 rolls in each; 12 ÷ 3 = 4.
3	20 muffins, 5 in each box	20 ÷ 5 = 4 (4 boxes).
4	3 × ___ = 18	6, because 3 × 6 = 18.
5	Division fact with 2, 7, and 14	14 ÷ 2 = 7 or 14 ÷ 7 = 2 (either is correct).

POST-LESSON CHALLENGE

#	Prompt	Model answer / grading note
1	Fact family for 6, 8, and 48	6 × 8 = 48, 8 × 6 = 48, 48 ÷ 6 = 8, 48 ÷ 8 = 6.
2	Unknowns: 7 × ___ = 63, ___ ÷ 4 = 9, 48 ÷ ___ = 6	9 (7 × 9 = 63); 36 (4 × 9 = 36); 8 (6 × 8 = 48).
3	6 trays of 6 paczki into boxes of 9	Step 1: 6 × 6 = 36 paczki. Step 2: 36 ÷ 9 = 4 boxes.
4	Mistake in 64 ÷ 8 = 7	He used a wrong fact (8 × 7 = 56, not 64). The quotient is 8, because 8 × 8 = 64.
5	Vocabulary match	dividend: the total you start with and split up; divisor: the number you divide by; quotient: the answer to a division problem; fact family: facts that use the same three numbers.

★ CHALLENGE

#	Prompt	Model answer / grading note
1	72 muffins in boxes of 8 or 9	Boxes of 8: 72 ÷ 8 = 9 boxes. Boxes of 9: 72 ÷ 9 = 8 boxes. Boxes of 9 use fewer boxes (8 instead of 9).